

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2022–23

Paper: CS-106 — Operating System Concepts

Time: Three Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions. All questions carry equal marks.***Question 1.**

[4 + 3 + 3 + 4 = 14]

- (a) Explain the different states of a process with the help of a state diagram.
- (b) Name any two differences between logical and physical addresses.
- (c) What is meant by pre-paging? Is it better than demand paging? Justify.
- (d) What will be the output of the following code? Draw the execution tree explaining all process creations.

```
1 #include <stdio.h>
2 #include <unistd.h>
3 int main() {
4     if (fork() || fork())
5         fork();
6     printf("1 ");
7     return 0;
8 }
```

Question 2.

[2 + 3 + 3 + 3 + 3 = 14]

- (a) List out the benefits and challenges of thread handling.
- (b) What is thrashing? How can this problem be resolved?
- (c) Explain the principle of locality.
- (d) Explain the working set model.
- (e) Explain Compaction with a suitable example.

Question 3.

[7 + 7 = 14]

- (a) Why are Translation Look-aside Buffers (TLBs) important? Explain the details stored in a TLB table entry.
- (b) Explain the concept of demand paging in detail with neat diagrams.

Question 4.

[4 + 10 = 14]

- (a) What are the different principles which must be considered while selecting a CPU scheduling algorithm? Support your answer with an example.

- (b) Consider the following set of processes with CPU burst times (in milliseconds):

Process	Arrival Time	Burst Time	Priority
P1	0	3	3
P2	1	5	1
P3	3	2	3
P4	9	5	2
P5	12	5	1

Draw a Gantt chart and find the average waiting time and average turnaround time for:

- (i) FCFS
- (ii) SJF (Preemptive)
- (iii) SJF (Non-Preemptive)
- (iv) Priority Scheduling (Preemptive; smaller number = higher priority)
- (v) Round Robin (Time Quantum = 1 ms)

Question 5.

[6 + 8 = 14]

- (a) Compare the following memory management techniques on their strengths and weaknesses: (i) Fixed Partition, (ii) Dynamic Partition, (iii) Simple Paging.
- (b) Given the following page reference string:
 1, 2, 3, 2, 5, 6, 3, 4, 6, 3, 7, 3, 1, 6, 3, 4, 5, 3, 2, 4, 3, 4, 5, 1
 Number of page frames = 4. Show the page trace and calculate the number of page faults for the following page replacement policies: (i) LRU, (ii) Optimal, (iii) FIFO. Also explain Belady's anomaly.

Question 6.

[8 + 6 = 14]

- (a) What is busy waiting with respect to process synchronization? Explain how a semaphore reduces the severity of this problem. Also define and distinguish between:
- (i) General (counting) semaphores
 - (ii) Binary semaphores
 - (iii) Strong semaphores
 - (iv) Weak semaphores
- (b) The following program consists of 3 concurrent processes (P0, P1, P2) and 3 binary semaphores initialized as S0=1, S1=0, S2=0:

<pre> /* P0 */ while(true){ wait(S0); print("0"); signal(S1); signal(S2); } </pre>	<pre> /* P1 */ while(true){ wait(S1); print("1"); signal(S0); } </pre>	<pre> /* P2 */ while(true){ wait(S2); print("2"); signal(S0); } </pre>
--	--	--

How many times will process P0 print '0'? Give step-by-step explanation.

Question 7. Write the Banker's Algorithm. Consider the following snapshot of a system at time T_0 with 5 processes (P0–P4) and 4 resource types (A, B, C, D): $3 + 4 + 3 + 4 = 14$

Process	Allocation				Maximum				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	0	0	1	2	0	0	1	2	1	5	2	0
P1	1	0	0	0	1	7	5	0				
P2	1	3	5	4	2	3	5	6				
P3	0	6	3	2	0	6	5	2				
P4	0	0	1	4	0	6	5	6				

Answer the following:

- What is the content of the NEED matrix?
- Is the system in a safe state? If yes, justify with step-wise calculations.
- Which processes may cause deadlock if the system is not safe?
- If a request from process P1 arrives for (0, 4, 3, 1) — can the request be granted immediately? Justify.